Causal Nowcasting of 10 ms Charge-Exchange Spectroscopy Gaps in KSTAR: an LSTM with Lower Error than Acausal Interpolation and the 20-Parameter Physical Filter It Computes

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Abstract

Charge-exchange spectroscopy (CES) measures ion temperature (T_i) and toroidal rotation (V_{rot}), but on the KSTAR 10 ms grid (641 discharges) 8.2% of T_i and 23.9% of V_{rot} samples are missing, and a further 41.1% of V_{rot} rows are instrument holds. Interpolation, the incumbent gap filler, reads the next measurement. We fill the gaps with a strictly causal two-branch LSTM that reads beam-emission spectroscopy, electron-cyclotron-emission imaging, Mirnov coils and the CES past, its V_{rot} branch routed away from the fast diagnostics. Two populations treat the 0.53% of T_i fit failures as missing (cut) or keep them (inclusive). *Ion temperature:* the skill over PCHIP interpolation is +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive), the 95% interval excluding zero on four held-out splits, and [+0.19, +0.25] / [+0.19, +0.31] in a repeated 5-fold replicate; in the cut population the margin comes from the fast diagnostics (windowed-control ablation), and a 20-parameter physical filter read off the network trained from scratch keeps 72–78% of its PCHIP-relative skill on 4/4 cut splits (bar 0.70). *Rotation:* the skill over PCHIP is [+0.07, +0.70] (cut) and [+0.05, +0.68] (inclusive) under a two-level bootstrap over eight new partitions, excluding zero, carried by the CES history alone: the fast channels correlate with dV_{rot}/dt at $r = -0.006 / -0.003$, and a physics filter's median error is 0.94–1.00× persistence, the 10 ms photon floor. The nowcaster's own median V_{rot} error is above persistence's (4.3 vs. 3.7 km/s); the aggregate skill of both targets is a tail statistic, and everything is one device and one campaign.

1. Introduction

Ion temperature T_i and toroidal rotation V_{rot} govern pedestal structure, ELM behaviour and access to high confinement. Both are measured by charge-exchange spectroscopy (CES) [1], which infers T_i from Doppler broadening and V_{rot} from the Doppler shift of the charge-exchange line. CES must integrate photons, so it is slow and intermittently missing: on the KSTAR 10 ms grid, 8.2% of T_i and 23.9% of V_{rot} observations are missing—independently of each other—and a further 41.1% of V_{rot} rows repeat the previous reading bit-for-bit, so only 35.0% of the grid carries an independent V_{rot} measurement. The missingness concentrates at low signal-to-noise phases, ELMs and transitions, i.e. it is missing-not-at-random [2]. Meanwhile beam-emission spectroscopy (BES), electron-cyclotron-emission imaging (ECEI) and Mirnov coils observe the same plasma without gaps, but none measures T_i or V_{rot}. We ask: *at a 10 ms step where CES is missing, can the simultaneous fast diagnostics plus the irregular CES past recover what interpolation of CES alone cannot?*

A deliberately hard bar. The comparators are *offline* interpolants of the CES channel (linear, monotone-cubic PCHIP [3], a Gaussian process) that read neighbours on both sides of the target, including the future; the model is strictly causal. A causal estimator with lower error than a future-using interpolant must draw on information that the CES time series alone does not contain. Because the strongest deployable competitor is a past-only smoother, a *causal* GP arm is reported separately.

Every result is reported in two co-primary populations: *cut*, in which the 0.53% of T_i rows above 3 keV (spectral-fit failures) are treated as missing in every arm, and *inclusive*, in which they are kept (§3).

Claims. The paper makes two claims, one per measured quantity.

1. **Ion temperature: a causal nowcast with lower error than acausal interpolation, and the equation it computes** (§4.5, §6.2–§6.4). The skill over future-using PCHIP is +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive) on four held-out splits and, as the overlap-corrected fold-level interval of a pre-registered repeated 5-fold replicate (100 runs), [+0.19, +0.25] / [+0.19, +0.31], excluding zero; over a causal GP it is +0.08 to +0.17 (interval above zero on 8/8 cells), and against the offline GP it is −0.05 to +0.11 (interval including zero on 7/8). In the cut population the margin comes from BES and ECEI (removing them drops the windowed control below PCHIP, paired −0.25 to −0.43, 4/4; in the inclusive population a history-only model keeps +0.15 to +0.23, a spike-robustness component). A four-stage pre-registered readout writes the T_i branch as a 20-parameter physical filter which, trained from scratch, keeps 72–78% of the backbone's PCHIP-relative skill on 4/4 cut splits (registered bar 0.70).
2. **Rotation: a lower mean error too, carried by the CES history alone** (§4.2, §6.3, §6.5). The skill over PCHIP, under the registered two-level bootstrap over eight new 5-fold partitions, is [+0.07, +0.70] (cut)

and [+0.05, +0.68] (inclusive), both excluding zero; the fold-level overlap-corrected interval [−0.02, +0.41] (cut) still includes zero. The fast channels correlate with dV_{rot}/dt at $r = -0.006$ (BES) and −0.003 (ECEI) against +0.070 and +0.078 with dT_i/dt; the 100 Hz Mirnov values are single raw samples (24/24 coils), and features of the raw 2 MHz Mirnov stream change the skill by −0.128 [−0.544, −0.105] (cut; six training and three test discharges). In the median a separate hold-or-jump physics filter has an error ratio to persistence of 0.94–1.00 (interval including 1 on 3/4 splits), the 10 ms photon floor, while the nowcaster's own median row is above persistence (4.3 vs. 3.7 km/s).

The protocol is pre-registered. The evaluation—two co-primary populations, a shot-clustered paired bootstrap, an overlap-corrected repeated cross-validation interval and a two-level bootstrap (§5)—is assembled from standard pieces of other fields and is described as method, not claimed. A full version of this paper (supplement) carries every ablation, sensitivity row and derivation omitted here.

2. Related work

Gap filling and baselines. GP regression [4] is the standard tool for fusion profile fitting [5, 6], applied to KSTAR CES T_i profiles [7]; PCHIP [3] avoids spline ringing at steps. We score with Murphy's skill [8]. Persistence as a reference has precedent in fusion profile forecasting [9] and space weather [10].

Neural diagnostic inference. NN-CES extracts T_i and rotation from KSTAR CES spectra without beam modulation [11]; it is the closest prior work and needs a measured spectrum at the step: *NN-CES reads the measured steps better; this work fills the ones that are not measured.* Lin et al. [12] map an x-ray crystal spectrometer to T_i and rotation on EAST, an instantaneous mapping from the same photon channel. RTCAKENN [13] reconstructs kinetic profiles in real time and tolerates dropped Thomson/CER inputs; there missingness is a robustness condition, here the target's own sparsity is the problem.

Cross-diagnostic reconstruction. Diag2Diag synthesizes Thomson T_e/n_e from simultaneous diagnostics on DIII-D [14] with a memory-less MLP; related work densifies Thomson on COMPASS [15], fills Thomson intervals causally with an extended Kalman filter on TCV [16], restores EAST T_e and Z_{eff} [17, 18] and masks channels on HL-3 [19]. These target electron or generic dense channels; we extend the programme to a sparse ion channel, to causal estimation conditioned on the target's own irregular past, and to per-target tests against pre-registered baselines. We found no precedent for that conjunction, and "not found" is not "does not exist".

Methods lineage. Observation masks follow GRU-D [20]; pre-registration follows registered reports [21]; the over-

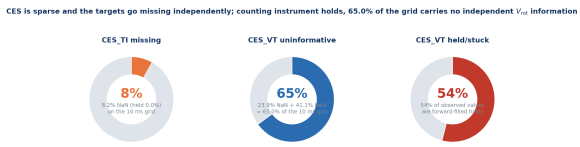


Figure 1: CES missingness on the 10 ms grid. T_i and V_{rot} go missing independently (8.2% / 23.9%); counting holds, 65.0% of the grid carries no independent V_{rot} information.

lap correction is [22]; the discharge-cluster bootstrap with training repeats as the outer level is a hierarchical bootstrap [23–25]; the equation retention test is that of [26] in non-inferiority form [27]; the Student-t measurement model is robust Kalman filtering [28], with tokamak precedent [29].

3. Data and problem

We use 641 KSTAR discharge files (shots 30801–32751) aligned to a 10 ms grid, 247,207 rows in total. Each row holds BES [30] (9 channels), ECEI [31] (4), Mirnov coils (2), time, and the two CES targets. T_i is missing in 8.2% of rows and V_{rot} in 23.9%, independently, so the loss is masked per target (Fig. 1). Rows group into contiguous segments at gaps ≥ 0.5 s; no arm bridges a gap.

Held V_{rot} values. 54% of observed V_{rot} values are bit-identical repeats of the previous observation (runs up to 1,214 rows, 499/641 files), because the native V_{rot} cadence is slower than the grid; T_i has 1 held row in 226,991. Holds are removed *everywhere*—targets, history inputs, normalization statistics and every baseline's anchors—so no arm scores a forward-fill for free.

Fit failures and two populations. 1,197 T_i rows (0.53%) exceed 3 keV (951 runs in 274 discharges, 85% of runs single rows), failed spectral fits that also poison interpolation anchors. We pre-registered **two co-primary populations**: *cut* (>3 keV treated as missing in every arm) and *inclusive* (no cut); a claim is stated without qualification only if it holds in both, in the form of non-inferiority trials with two analysis sets [32, 33]. The threshold is immaterial across 2.5–4 keV (mean T_i skill +0.230 and +0.232 against +0.236 at 3 keV).

4. Model

The main model, `seq_v2`, is a 357,570-parameter causal sequence nowcaster. This section fixes what is estimated, why the inputs are wired to each target, which time scales the model sits on, and what equation the trained network computes. The full version (supplement) gives in addition the full windowed-control architecture, the interpretable rung, the optimization setup in detail, and the stage-by-stage readout with its intermediate parameter bands.

4.1. Estimand, loss and an irreducible floor

At grid time t the model estimates the normalized $y_t = [T_i, V_{\mathrm{rot}}]_t$ from $x_{\le t}$ with per-target observation masks $m_{t,c} \in$

Table 1: Term-by-term audit of (3) (641-shot census).

Term	In our data?	Disposition
$\partial L/\partial t$	consecutive observed pairs	measured
$T_{\rm NBI}$ (beam torque)	no 0-D channel exists	absent
$T_{\rm NTV} \sim \delta B^2$	Mirnov, decimated to 100 Hz	closed negative
$\nabla\cdot\Pi_\varphi$ (turb. flux)	BES density fluctuation	measured, $r \approx 0$
$T_{\rm int}$ (residual stress)	needs ∇T_i; scalar held	not reachable

$\{0, 1\}$:

$$\mathcal{L} = \frac{\sum_{t,c} m_{t,c}\,(\hat{y}_{t,c} - y_{t,c})^2}{\sum_{t,c} m_{t,c}} + \lambda\,\overline{\mathrm{ReLU}(z_0 - \hat{y}_{t,T_i})}, \qquad \lambda = 0.1, \quad (1)$$

whose first-term minimizer is $\mathbb{E}[y_t \mid x_{\le t}]$; the second term discourages T_i below the normalized zero z_0. A row with one observed target still supervises it (a both-required filter would discard ≈28% of labelled rows). The target is itself a photon-integrated fit with noise $\sigma_{\rm meas}$, so even a perfect physical estimate scores $\mathrm{RMSE}^2 = \sigma^2_{\rm meas} + b^2$. Difference estimators put $\sigma_{\rm meas}$ for T_i (cut) at 46.4–129.9 eV, the order of the backbone's 157.8 eV, and 46.6% of the fourth-difference squared mass (65.6% for $V_{\rm rot}$) sits in the top 1% of one-step changes: (1) is a *tail statistic* (§6.5).

4.2. Two conservation laws fix the input routing

Ion energy and toroidal angular momentum obey

$$\partial_t\left(\tfrac{3}{2} n_i T_i\right) + \nabla\cdot\mathbf{q}_i = Q_{ei}\,(T_e - T_i) + S_i, \quad (2)$$

$$\partial_t\left(n_i m_i \langle R^2\rangle \omega_\varphi\right) + \nabla\cdot\Pi_\varphi = T_{\rm NBI} + T_{\rm NTV} + T_{\rm int} + \cdots \quad (3)$$

Both constrain a *rate*, so the question is whether the fast diagnostics correlate with $dV_{\rm rot}/dt$. Table 1 checks each term of (3) against the channels we hold. The one reachable term, the turbulent momentum flux that BES sees through the density fluctuation, measures

$$r(\mathrm{BES}, dV_{\rm rot}/dt) = -0.006, \quad r(\mathrm{ECEI}, dV_{\rm rot}/dt) = -0.003,$$

against +0.070 and +0.078 for dT_i/dt in the energy equation (levels: T_i +0.341, +0.311; $V_{\rm rot}$ +0.027, +0.005). Because the same measurement finds the energy-equation coupling, the rotation result is not a limitation of the method. With $T_{\rm NBI}$ absent, $T_{\rm NTV}$ closed and $T_{\rm int}$ unreachable, no route exists from our inputs to $V_{\rm rot}$ through (3), so the branches are split at the encoder. For T_i the channel is collisional electron–ion coupling $Q_{ei}(T_e - T_i)$: ECEI supplies T_e, BES n_e structure.

4.3. Time scales

The grid spacing $\Delta t = 10$ ms gives $f_{\rm Nyq} = 1/(2\Delta t) = 50$ Hz. Mode rotation sits in the kHz band (edge harmonic oscillations near 4 and 8 kHz in this dataset), so a dB/dt stream decimated to 100 Hz without anti-aliasing cannot carry it; the Mirnov lag-1 autocorrelation of −0.009 (BES +0.568) confirms it. The electron–ion equilibration time $\tau_{\rm eq} \simeq (m_i/2m_e)\,\tau_e$, with the Braginskii $\tau_e = 3.44{\times}10^5\,T_e^{3/2}/(Z\,n_e \ln\Lambda)$, spans 8.3–305 ms over $T_e = 0.5$–3 keV and $n_e = 2$–$5{\times}10^{19}$ m^{-3} and 8–59 ms near the observed T_i median of 593 eV; the measured saturation of causal context is about 50 ms, the same order (an order-of-magnitude consistency, not a measurement of n_e, T_e). Autocorrelation $1/e$ times agree for T_i (159 ms), BES (161) and ECEI (147), whereas $V_{\rm rot}$ gives 16 ms without holds and more than 300 ms with them: its relaxation time cannot be measured in this dataset.

4.4. Full-grid causal sequence backbone

`seq_v2` keeps every input-complete row of a contiguous segment, labelled or not, as context, because the fast diagnostics have no gaps (Fig. 2; cell in Fig. 3). Each step carries 22 strictly causal channels: the 15 fast channels, standardized per discharge by that discharge's own mean and variance; $\log(1{+}\Delta t_{\rm prev})$; and per target the last observed value carried forward from rows $< t$, log(1+staleness) and a has-any-past-observation flag. The T_i branch reads all of them; the $V_{\rm rot}$ branch reads only the seven non-fast channels—the implementation of §4.2. Because the split is at the encoder, no shared recurrent state carries fast information into the $V_{\rm rot}$ head. Outputs are normalized $[T_i, V_{\rm rot}]$ per row, trained with (1) over every labelled row (AdamW, $\eta = 10^{-3}$, weight decay 10^{-4}, batch 16 segments, clipping 1.0, early stopping with a 30-epoch cap). Splits are file-level, normalization is train-file-only, and the target row never enters the input. The model's reach into the past is the whole segment rather than a fixed window.

Windowed control and the gate. The control (201,258 parameters) encodes each modality with its own time-aware 1-D CNN over the W=2 preceding rows and reads the CES history with a bidirectional GRU pooled by one multi-head attention readout per target, hard-masked so that attention mass lands only on rows where that target was measured; its $V_{\rm rot}$ head also fuses history and time only. The full-grid framing was adopted over it by a pre-registered four-condition gate on a 4×4 grid of split × initialisation seeds: paired T_i skill positive on 16/16 runs, pooled +0.081 [+0.067, +0.096]. What the framing buys is reach: the control's skill over the causal GP is +0.041 (interval including zero on 3/4 splits, cut), the backbone's +0.08 to +0.17.

Size, family and cost. An interpretable rung that starts at persistence and adds a bounded tanh latent (21,498 parameters, 6% of the backbone) has a paired T_i skill against the backbone of mean +0.002 under the cut (every interval containing zero) and −0.194 in the inclusive population. Widening the T_i encoder over a 26× parameter range gives mean T_i skill +0.230 to +0.236 (cut); averaging four initialisations adds a paired +0.020, on the pre-registered 0.02 band. Context saturates at 5 steps = 50 ms, where the skill over the

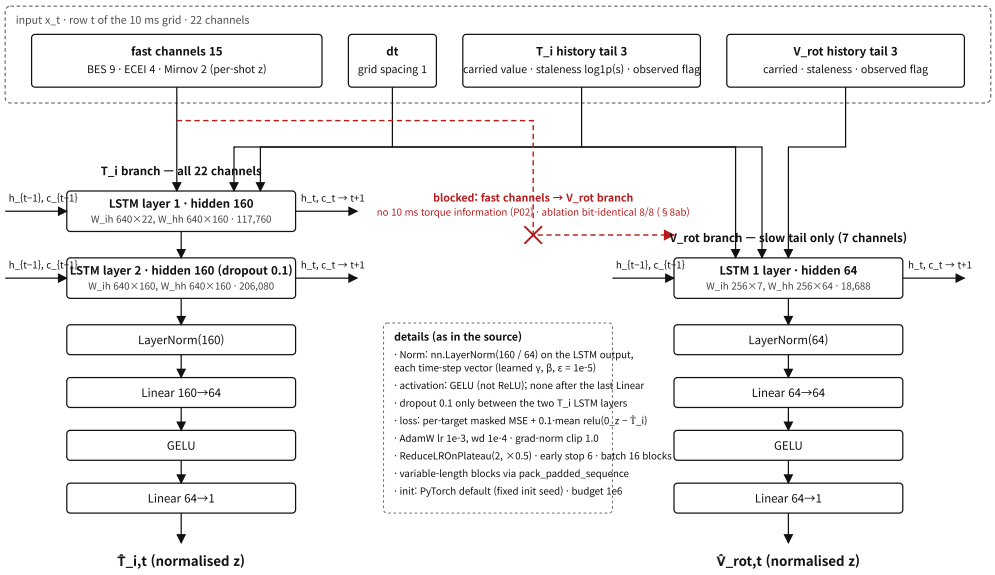


Figure 2: `seq_v2` for one grid row t. 22 inputs (15 fast, per-shot z; Δt; per target carried value, staleness, observed flag) feed two causal LSTMs. The T_i branch reads all 22 (2 layers, 160 units), the $V_{\rm rot}$ branch only the seven slow channels (1 layer, 64 units); each ends in LayerNorm and Linear–GELU–Linear. The dashed red path from the fast channels to $V_{\rm rot}$ does not exist by construction. 357,570 parameters.

causal GP is +0.104 [+0.079, +0.128]. The batch of 16 is checked against the gradient noise scale computed exactly over the 452 training blocks,

$$B_{\rm simple} = \frac{\operatorname{tr}\Sigma}{\|G\|^2}, \qquad G = \tfrac{1}{N}\textstyle\sum_i g_i,$$
$$\operatorname{tr}\Sigma = \tfrac{1}{N}\textstyle\sum_i \|g_i\|^2 - \|G\|^2, \tag{4}$$

whose post-initial median is 33.6 blocks (range 6.0–61.4), so batch 16 sits at about half the noise scale; after the first epoch 86–98% of a single block's gradient energy is noise. A stateful step costs 0.97 ms median and 1.49 ms at p99 on CPU.

4.5. The equation the network computes

A pre-registered readout of the thirty frozen `seq_v2` checkpoints found that weight magnitude carries no stable information (first-layer column norms are flat across the fifteen fast channels and per-channel ranks do not reproduce across splits), whereas the input–output response does: the T_i output is an exponentially weighted average of past observed T_i plus a same-instant response to the fast channels, with a forget-gate memory of 1.3 grid steps. The readout ran in four registered stages. *(1) Frozen rows:* an unfitted running mean carries 73–82% (cut) of the backbone's skill over persistence, and a ridge-fitted linear term in the 15 ten-millisecond increments raises it to 77–85% (4/4 splits above the 0.75 bar). *(2) Trained increment form:* the same equation trained from scratch keeps 50–66% of the PCHIP-relative skill, below the 0.70 bar, and a cross term adds less than 0.001. *(3) Jacobian:* on every frozen TEST row the backbone's fast-channel response is a ≈100 ms high-pass (current value minus the preceding ten-step mean) with gain flat to 2σ and switched off beyond 3σ, independent of the recent increments—the form, not a missing state-dependent term, was the gap. *(4) Corrected form*, trained from scratch under the identical protocol:

$$\hat{y}_t = m_t + \sum_{c=1}^{15} a_c\, d_{c,t}\, \sigma\!\left(\frac{\theta - |d_{c,t}|}{0.5}\right) + b_0, \tag{5}$$

$$d_{c,t} = x_{c,t} - \bar{x}_{c,\,t-10..t-1},$$
$$m_t = m_{t-1} + K(s)\,(y^{\rm obs} - m_{t-1}),$$
$$K(s) = K_\infty + (K_0 - K_\infty)\, e^{-s/s_K}, \tag{6}$$

where m_t is updated only when a new T_i measurement arrives, s is the gap since the previous one, and $\sigma(\cdot)$ is the logistic function, with $d_{c,t}$ and θ in per-discharge standard deviations; the first observation of a segment is taken whole ($K = 1$). Its 20 parameters are three gain constants, one gain a_c per fast channel, the threshold θ and the offset b_0. Across the four cut splits the learned values are $K_0 = 0.27$–0.30 for a fresh measurement, $K_\infty = 0.48$–0.53 after a long gap, transition $s_K = 39$–47 ms, $\theta = 2.9$–3.0σ, $\sum a_{\rm BES} = +48$ to $+53$ and $\sum a_{\rm ECEI} = +44$ to $+50$ eV per σ. Read physically: a

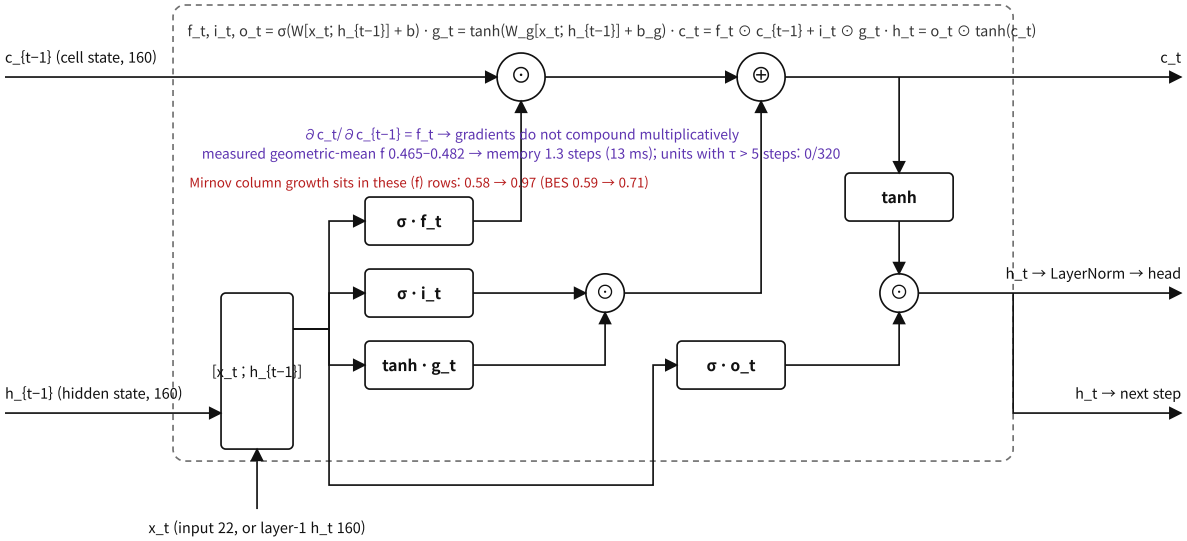


Figure 3: One LSTM cell (T_i branch, layer 1, 160 units): $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, so $\partial c_t/\partial c_{t-1} = f_t$ and the memory length is set by f. Purple: values measured on the frozen weights (geometric-mean f 0.465–0.482, memory 1.3 steps; 0 of 320 units with $\tau > 5$ steps). Red: where the Mirnov input columns grow during training (forget-gate rows, 0.58 → 0.97 against 0.59 → 0.71 for BES).

Kalman-like observation filter that trusts a new CES sample more the longer it has waited, plus a linear response to each fast channel's departure from its 100 ms baseline that rejects outliers beyond 3σ. Its retention of the backbone's skill is in §6.4. These are measurements of the model; the headline numbers remain the backbone's.

5. Evaluation protocol

Metric and baselines. Errors are computed per target in physical units, and the primary score is $\text{skill}_{\text{PCHIP}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{PCHIP}}$ (0 = equal error). Every arm is scored on the identical (file, row) set. Persistence and a local AR are causal; linear, PCHIP and an offline GP (Matérn-3/2 + white noise, nearest 16+16 neighbours, per-sample marginal-likelihood selection) read the future; the causal GP is the same GP on the 16 past neighbours. Where no future neighbour exists the interpolant predicts persistence (0.3–0.4% of scored T_i rows, but 40–44% of V_{rot} rows, so "vs. PCHIP" for V_{rot} is two-fifths "vs. persistence").

Splits and pre-registration. Files are split train/val/test with four seeds (42, 1, 7, 123); test files were reserved before any architecture search and are never read during selection. The canonical test set holds 32,589 T_i rows over 96 shots and 10,463 V_{rot} rows over 60 shots (cut). "Pre-registered" means the hypotheses, decision rules and numerical predictions of a batch were committed to the repository before it was run; there is no third-party registry.

Intervals. Per-sample paired squared-error differences are aggregated by shot and whole shots resampled (10,000 resamples); a result is reported with whether its 95% interval excludes zero. Because the four splits share most of their training discharges, the backbone was re-run as shot-level repeated 5-fold cross-validation on 639 discharges (ten repeats, 100 runs, every discharge tested once per repeat). *Fold-level*: the Nadeau–Bengio interval $\bar{S} \pm t_{0.975,n-1}\sqrt{(1/n + n_{\text{test}}/n_{\text{train}})\, s^2}$ [22]. *Pooled*: each repeat's five test folds concatenated into one skill, with a two-level bootstrap that draws a repeat and then resamples its discharges, over the eight repeats run after the rule was registered; this pooled readout was named the deciding one. Crossed resampling overestimates variance [34, 35], so the interval is conservative.

6. Results

6.1. RMSE ladder

Table 2 gives the physical-unit RMSE on the canonical test split. Among the methods that do not read the future, `seq_v2` has the lowest RMSE, 4% (T_i) and 18% (V_{rot}) below the causal GP, and it is also below linear and PCHIP interpolation. The offline GP, which averages noisy anchors, is the one arm with a lower T_i RMSE (153.8 vs. 157.8).

6.2. Headline and replicate

T_i. Table 3: $\text{skill}_{\text{PCHIP}}$ = +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive), every shot-clustered interval excluding

Table 2: RMSE, seed-42 test split, cut population, genuine measurements only (n = 32,589 T_i, 10,463 $V_{\rm rot}$). Lower rows read the future. Inclusive: `seq_v2` 363.0 / 23.7, PCHIP 412.4 / 30.2, causal GP 394.6 / 28.8, persistence 478.0 / 33.4.

Arm	Information	T_i (eV)	$V_{\rm rot}$ (km/s)
`seq_v2`	fast + past CES, segment	**157.8**	**23.6**
Windowed control	fast + 2 rows past CES	169.2	26.1
Causal GP	past 16 CES	164.3	28.8
Persistence	last observed CES	197.2	33.4
AR (local)	past CES	472.2	51.0
GP (offline)	past + future CES	153.8	24.7
Linear interp.	past + future CES	169.8	29.0
PCHIP	past + future CES	173.6	30.2

zero on all four splits; over the causal GP +0.08 to +0.17 and over persistence +0.36 to +0.46, both excluding zero on 8/8 population–split cells. Against the offline GP the paired skill is −0.05 to +0.11 with the interval including zero on 7/8 cells, so "lower error than future-using interpolation" refers to the pre-registered interpolants, not to every offline smoother. The replicate reproduces every verdict: vs. PCHIP the fold-level interval is [+0.19, +0.25] (cut) and [+0.19, +0.31] (inclusive), the pooled one [+0.19, +0.25] and [+0.19, +0.29]. The skill is not confined to adjacent history: beyond $\Delta t > 15$ ms since the last observation it is +0.268 (cut) and +0.206 (inclusive) over PCHIP, both intervals excluding zero.

$V_{\rm rot}$**.** Point estimates are positive on every split, but the four-split interval excludes zero on 1/4 (cut) and 2/4 (inclusive) splits, which alone supports no rotation claim. Scored discharge by discharge across eight fresh partitions, the registered pooled two-level interval over PCHIP is [+0.07, +0.70] (cut) and [+0.05, +0.68] (inclusive), over persistence [+0.28, +0.76] and [+0.14, +0.75], and over the causal GP [+0.05, +0.70] and [+0.02, +0.68], all excluding zero; the fold-level interval still includes zero ([−0.02, +0.41] cut) because an 80-discharge fold's ratio is set by a few heavy-error discharges.

Why inclusive numbers are higher. Every arm scores higher against PCHIP without the cut (backbone mean +0.268 vs. +0.236) because spikes poison the interpolation anchors, which a learned model can discount; this is why neither population is quoted alone.

Two stress tests. *Missing-not-at-random:* reweighting the scored rows onto the covariate distribution ($\Delta t \times$ local activity) of the genuinely missing, in-domain rows—54–68% of missing T_i rows—gives T_i skill over persistence +0.28 to +0.44 with intervals excluding zero on 8/8 cells, and over PCHIP +0.14 to +0.28 with intervals excluding zero on 2/4 cut and 4/4 inclusive splits (missing $V_{\rm rot}$ rows are only 4–6% in-domain, and we draw nothing from them). *Campaign shift:* with a strictly temporal split (train 416 shots, val 128, test the 97 latest), `seq_v2` keeps +0.17 to +0.20 over PCHIP and +0.11 to +0.16 over the causal GP, intervals excluding zero on 4/4 initialisations in both populations, where the windowed control keeps −0.001 to +0.091 (2/4 cut, 0/4 inclusive). The fast diagnostics drift 1.22σ (BES) and 0.53σ (ECEI) between periods against 0.115σ for the targets; `seq_v2` standardizes each discharge by its own statistics.

6.3. The $T_i \leftrightarrow V_{\rm rot}$ asymmetry

Zeroing one input group at evaluation time on the windowed control (Table 4) separates the two targets. History is indispensable for both. For T_i in the cut population, removing the fast channels drops the skill below the interpolator (paired −0.25 to −0.43, 4/4 significant): the margin over interpolation is fast-diagnostic information. In the inclusive population the history-only model still has +0.15 to +0.23 over PCHIP, a spike-robustness component, which is why the cut population is the one that isolates the fast contribution. For $V_{\rm rot}$ the output is bit-identical on 8/8 cells without the fast channels; that is a consequence of the routing, not evidence for it. The evidence is the rate correlation of §4.2 (−0.006 / −0.003 for $dV_{\rm rot}/dt$ against +0.070 / +0.078 for dT_i/dt) and the two direct tests below.

Two routes to rotation information, tested. *Mirnov:* on 24 of 24 coils checked against the raw 2 MHz stream (12 discharges), the 100 Hz value equals a single raw sample at the grid time (lag 0), a point snapshot with no averaging or filter, so the kHz mode rotation is aliased away. Fifteen registered causal 10 ms features of the raw stream (RMS envelope, band powers, cross-spectral peak) routed into the $V_{\rm rot}$ branch change its skill by −0.128 [−0.544, −0.105] (cut) and −0.164 [−0.473, −0.133] (inclusive), trained on six discharges and scored on three $V_{\rm rot}$-carrying test discharges, 0 of 3 improving, although the coil physics is intact (the edge harmonic oscillation of #32092 appears at 4.39 and 8.91 kHz); within a discharge the dominant mode frequency correlates with $V_{\rm rot}$ at $|\rho| \le 0.11$ in six of seven discharges. *Beam torque:* no torque channel exists, and between shots the ECE T_e surrogate correlates with T_i at $r = +0.353$ but with $V_{\rm rot}$ at $r = +0.024$. NBI torque sets the rotation *level*, but beam power is a step input: the part of the 10 ms increment variance carried by a 300 ms moving average is 0.34% for $V_{\rm rot}$ (0.25% for T_i), so a perfect torque channel would matter for forecasts of 100 ms and longer [36], not for the 10 ms nowcast, whose level is already in the input as the value observed 10 ms earlier. The statement is scoped to standard CES at 10 ms from the core to the pedestal top; edge rotation inside an ELM cycle [37] is a different measurement.

6.4. Equation retention

Trained from scratch under the identical protocol, the 20-parameter equation (5)–(6) keeps **72–78%** of the backbone's PCHIP-relative skill and 87–90% of its persistence-relative

Table 3: Skill of `seq_v2` against each comparator. *Four splits*: range of the per-split skill over seeds 42/1/7/123 with the number of splits whose shot-clustered 95% CI excludes zero. *Fold-level*: Nadeau–Bengio 95% interval over 50 folds of repeated 5-fold CV, with the number of folds whose own CI excludes zero. *Pooled*: two-level bootstrap 95% interval over the eight registered repeats. Bold = interval excludes zero. The four-split T_i vs. offline GP entry pools both populations; "–" = not in the full version.

		Four splits		Fold-level (repeated 5-fold)		Pooled (two-level)	
Target	vs.	cut	inclusive	cut	inclusive	cut	inclusive
T_i	PCHIP	+0.174 to +0.264 (**4/4**)	+0.225 to +0.316 (**4/4**)	[**+0.186**, **+0.253**] (50/50)	[**+0.188**, **+0.307**] (46/50)	[**+0.188**, **+0.248**]	[**+0.192**, **+0.294**]
	persistence	+0.360 to +0.415 (**4/4**)	+0.406 to +0.459 (**4/4**)	[**+0.368**, **+0.410**] (50/50)	[**+0.361**, **+0.445**] (50/50)	[**+0.369**, **+0.408**]	[**+0.364**, **+0.438**]
	causal GP	+0.078 to +0.138 (**4/4**)	+0.123 to +0.169 (**4/4**)	[**+0.084**, **+0.138**] (47/50)	[**+0.067**, **+0.170**] (40/50)	[**+0.085**, **+0.132**]	[**+0.076**, **+0.157**]
	offline GP	−0.05 to +0.11 (CI includes 0 on 7/8)		[−0.044, +0.020] (1/50)	[−0.026, +0.094] (12/50)	[−0.043, +0.014]	[−0.017, +0.075]
$V_{\rm rot}$	PCHIP	+0.135 to +0.390 (1/4)	+0.132 to +0.384 (2/4)	[−0.021, +0.405] (15/50)	[−0.053, +0.391] (11/50)	[**+0.072**, **+0.701**]	[**+0.046**, **+0.683**]
	persistence	+0.30 to +0.50 (3/4 in each)		[**+0.190**, **+0.601**] (42/50)	[**+0.159**, **+0.587**] (38/50)	[**+0.277**, **+0.763**]	[**+0.139**, **+0.747**]
	causal GP	+0.020 to +0.331 (2/4)	+0.018 to +0.324 (2/4)	[−0.060, +0.512] (11/50)	[−0.096, +0.502] (10/50)	[**+0.045**, **+0.701**]	[**+0.023**, **+0.683**]
	offline GP	–	–	[−0.234, +0.312] (0/50)	[−0.268, +0.300] (0/50)	[−0.119, +0.635]	[−0.139, +0.609]

Table 4: Evaluation-time modality ablation of the windowed control: $\text{skill}_{\text{PCHIP}}$ (4-split mean) and the paired change against the full model (* = significant splits).

Inputs		cut	inclusive
Full	T_i	+0.173	+0.238
No fast	T_i	−0.125; −0.25 to −0.43 (4/4*)	+0.201; −0.03 to −0.09 (2/4*)
No history	T_i	−2.11	−1.16
Full	$V_{\rm rot}$	+0.213	+0.206
No fast	$V_{\rm rot}$	+0.213; +0.000 × 4 (bit-id.)	+0.206; +0.000 × 4 (bit-id.)
No history	$V_{\rm rot}$	−2.89	−3.51

skill on 4/4 cut splits, above the registered bar of 0.70, and sits above the increment-form equation of stage 2 by a paired +0.034 to +0.042 with intervals above zero on 4/4 splits. The remaining 22–28% is the network's nonlinear event handling, which no low-order function of the fast-channel history reproduces. The physical bands are read from the cut population only: in the inclusive population the equation keeps 0.43–0.81 of the PCHIP-relative skill and its BES gain changes sign under the fit-failure spikes.

6.5. Mean and median: a dual readout

An MSE skill is decided by few rows: the backbone's top 1% of T_i squared-error rows carry 48% (cut) and 83% (inclusive) of the total. On a typical row the backbone carries its anchor: its median absolute T_i error of 47.2 eV equals persistence's 46.1 eV and the target's own 10 ms scatter of 46.4 eV, while PCHIP reaches 39.4 eV by averaging with the future neighbour (median-AE skill −0.198 [−0.259, −0.135], win rate 0.48); the error ratio model/PCHIP falls below 1 only in the upper error quartile. For $V_{\rm rot}$ the backbone's median row is worse than persistence (4.3 vs. 3.7 km/s; median-AE skill vs. PCHIP −0.488 cut and −0.518 inclusive, interval below zero on 4/4 splits): its aggregate skill is a bulk deficit offset by a tail gain. The median is therefore read from a separate, pre-registered estimator: a scalar hold-or-jump filter per target run causally on the CES history, whose prior holds the level and lets it jump, with Student-t ($\nu_0 = 3$) measurement noise at the 10 ms bulk scatter. Its median absolute error is 0.911–0.920× (cut) and 0.906–0.915× (inclusive) persistence for T_i, median skill over persistence +0.08 to +0.09 with intervals excluding zero on 4/4 splits in both populations (8/8 split–population cells). For $V_{\rm rot}$ the filter's ratio is 0.94–1.00× with the interval including 1 on 3/4 splits, also after fitting the filter on TRAIN, while its median skill over the backbone is +0.09 to +0.27 (4/4 in both populations). The reason is measured: the ratio of relative T_i scatter to Doppler velocity scatter is the $\sqrt{2}$ of a Gaussian line fit in 8/8 split–population cells, so the 10 ms scatter sits at the photon floor, and between labels the median-optimal point is the last value. The filter's MSE skill against the backbone is −0.06 to −0.52, because event rows stay late: median from the filter, mean from the nowcaster. A Student-t predictive head on the backbone gives a calibrated uncertainty (cut: 50% / 90% coverage 0.48–0.51 / 0.91–0.92; in the inclusive population only a two-component mixture head passes the guards, coverage 0.53–0.55 / 0.90–0.91).

7. Limitations and conclusion

Limitations. The aggregate skill is a tail statistic: on the median row the backbone's T_i error equals persistence's and PCHIP's is 20% lower, and its $V_{\rm rot}$ median error is above persistence's; the median is reported from the physics filter beside, not inside, the aggregate. The unit of replication is the shot (96 T_i / 60–66 $V_{\rm rot}$ test shots per split), which bounds power. Skill is measured on observed points; the reweighting reaches 54–68% of missing T_i rows and 4–6% of missing $V_{\rm rot}$ rows. The offline claim is bounded above by the offline GP. The value cut is a one-sided proxy for fit quality; per-exposure fit metadata, which NN-CES [11] can provide, would replace it. Campaign transfer rests on one temporal block with four initialisations. Predictive coverage is reported over pooled rows, not per discharge. Everything is one device and one campaign.

Conclusion. A missing 10 ms CES row in KSTAR can be filled by a strictly causal estimator with lower mean-squared error than interpolation that reads the future, for ion temperature and for rotation; what separates the two quantities

is what the fast diagnostics contain. *Ion temperature:* the skill over PCHIP is +0.17 to +0.26 (cut) and +0.23 to +0.32 (inclusive) on four splits, [+0.19, +0.25] and [+0.19, +0.31] as the overlap-corrected fold-level interval of a repeated 5-fold replicate, and +0.08 to +0.17 over a causal GP on 8/8 cells, against −0.05 to +0.11 over the offline GP (interval including zero on 7/8). In the cut population the margin comes from BES and ECEI; the network computes a 20-parameter physical filter—a gap-dependent observation gain plus an outlier-rejecting 100 ms high-pass response to each fast channel—that keeps 72–78% of its PCHIP-relative skill. In the median a hold-or-jump physics filter reaches 0.906–0.920× persistence on 8/8 cells. *Rotation:* the skill over PCHIP under the two-level bootstrap is [+0.07, +0.70] (cut) and [+0.05, +0.68] (inclusive), excluding zero, carried by the CES history alone: the fast channels correlate with dV_{rot}/dt at $r = -0.006$ and -0.003, the 100 Hz Mirnov values are single raw samples, and raw 2 MHz features change the skill by −0.128 [−0.544, −0.105] (cut, three test discharges). Beam power is a step input whose band slower than 300 ms carries 0.34% of the 10 ms V_{rot} increment variance, so a torque channel is a lever for forecasts of 100 ms and longer. In the median the physics filter's V_{rot} error is 0.94–1.00× persistence (interval including 1 on 3/4 splits), the 10 ms photon floor. The aggregate skill of both targets is a tail statistic, and the statement is scoped to standard CES at 10 ms from the core to the pedestal top. The lever that remains for the 10 ms nowcast is per-exposure fit quality from the CES analysis. The full version (supplement) gives the gap-regime, peak-stratum, context and family ladders, the loss and distribution-head experiments and the complete readout. Code, pre-registrations and evaluation artifacts: https://github.com/iseungsang01/ml-intern-revision.

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